

[Total No. of Questions - 9] [Total No. of Printed Pages - 2]

Dec-24-0343 NEP

MCA-7303 (Cloud Computing)

MCA-3rd CBCS/NEP

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note: Attempt five questions in all, select one question each from section A, B, C, D. Section E (Question-9) is compulsory.

Section -A

1. Draw and explain the NIST cloud computing reference architecture. (12)
2. (a) What is Virtualization? Explain its different types. (6)
(b) What are Hypervisors? Explain with diagram its two basic types (6)

Section -B

3. (a) Draw and explain Map Reduce model with proper example. (6)
(b) Explain Google File System (GFS) architecture in detail. (6)
4. (a) What is DBaaS? Explain in detail. (6)
(b) Explain with proper diagram deployment models in Cloud Computing. (6)

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Section -C

5. (a) Write down the security fundamentals in Cloud. (6)
(b) What do you understand by shared security model in cloud computing? (6)
6. Differentiate between (any two): (6×2=12)
 - i) Relational and Non-Relational databases.
 - ii) Distributed Computing and Cloud Computing
 - iii) Public and Private Cloud.

Section -D

7. Differentiate between Dynamo, Hbase and Big table. (12)
8. Name and explain actors in DMTF reference architecture. (12)

Section-E (Compulsory)

9. (a) Define parallel Computing. (2)
(b) Name any two services provided by Google App Engine. (2)
(c) What is XaaS? (2)
(d) What is Community Cloud? (2)
(e) Define Multitenancy. (2)
(f) Name three core capabilities of Cloud Security. (2)